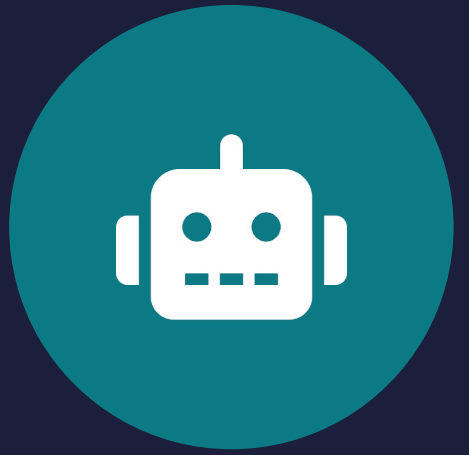
Introduction to Artificial Intelligence



Lecture 1 — What is AI, and how did we get here?

Master in Business Analytics & AI · Year 1

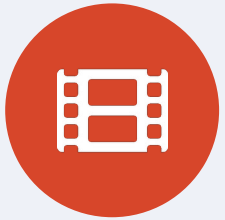
Warm-up: which of these is AI?



Calculator



Spam filter



Netflix recommendations



Smart thermostat



ChatGPT



Chess engine

So what is AI?

AI is the field that builds machines that can **act effectively** in **new situations**.

Based on Russell & Norvig, Artificial Intelligence: A Modern Approach, ch. 1



Some ask: does it behave like a human?

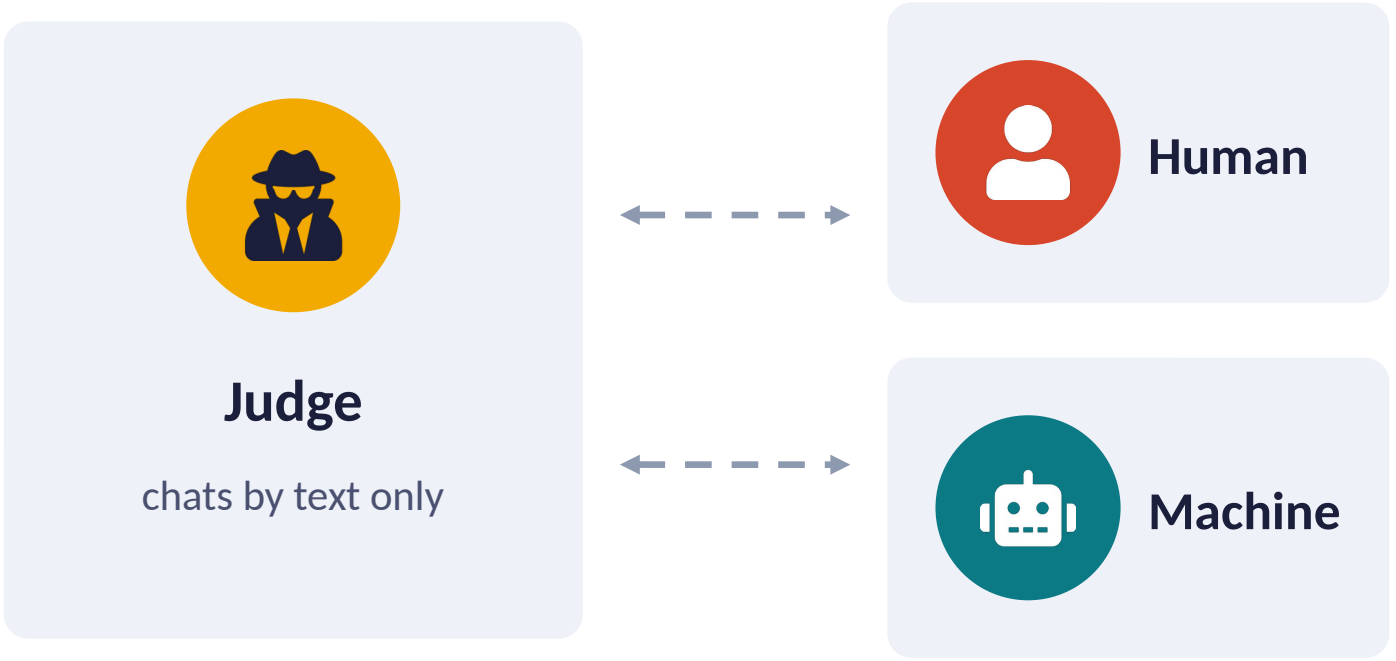


Others ask: does it do the right thing?

Four ways to define AI

	Like a human	Rationally (the "right thing")
Thinking	**Thinking humanly** Model how the mind works (cognitive science)	**Thinking rationally** Reason with logic — "laws of thought"
Acting	**Acting humanly** Behave like a person — the Turing test	**Acting rationally** Choose the action with the best expected outcome ★ Most of today's AI

Acting humanly: the Turing test (1950)



To pass, a machine needs:

- Language understanding
- Knowledge
- Reasoning
- Learning
- *+ vision & robotics for the "total" test*

Does ChatGPT pass? Does that make it intelligent?

Thinking like a human, or thinking logically?



Thinking humanly

Build models of how the mind works, then test them against how people actually think.

Field: cognitive science

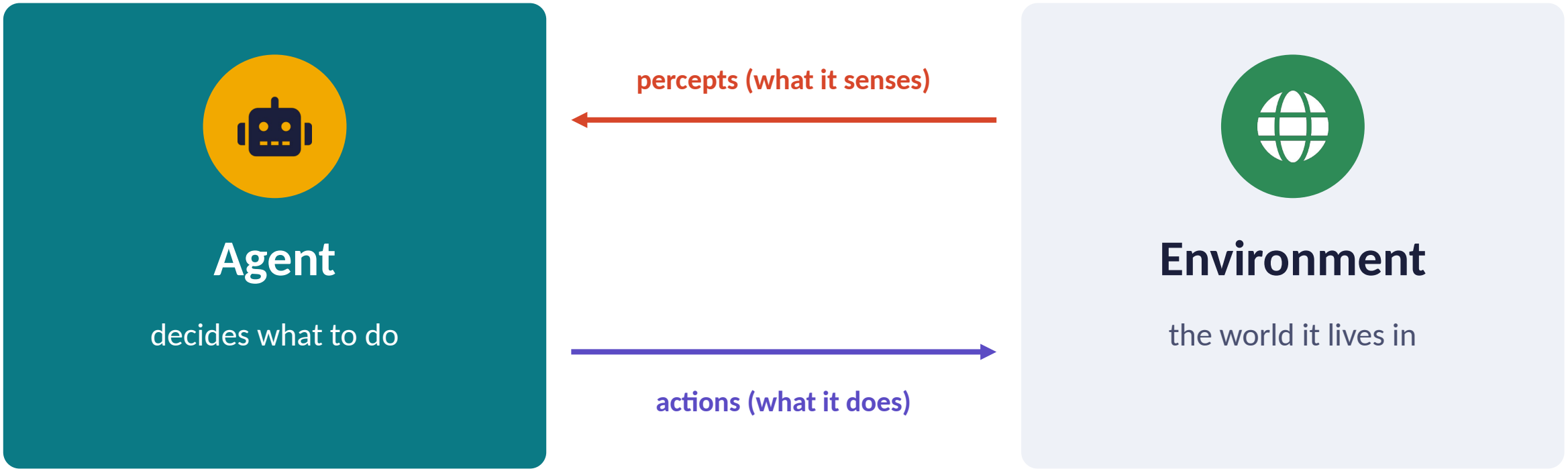


Thinking rationally

Use logic to reach correct conclusions. "All humans are mortal; Socrates is a human; so..."

Problem: the real world is uncertain

Acting rationally: the agent



Rational = choose the action with the best expected result, given what you know.

But who decides what is “best”?



The King Midas problem

He asked for everything he touched to turn to gold — and got exactly that. Including his food.



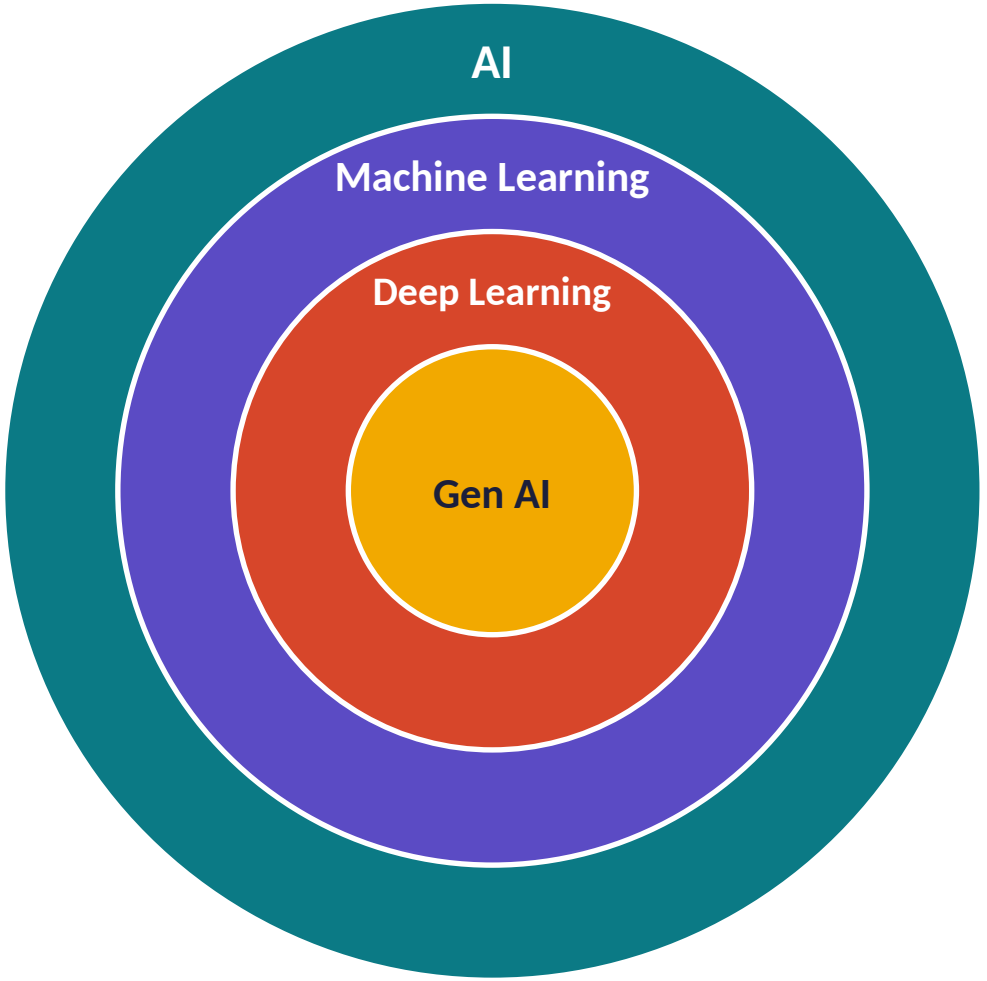
Today’s version

Tell a recommender to “maximise clicks” and it may learn to push outrage and clickbait. It did exactly what we asked — not what we wanted.

AI, Machine Learning, Deep Learning

Three terms people mix up — and how they fit together

How the terms fit together



Artificial Intelligence

Machines that act intelligently — with or without learning

Machine Learning

Systems that improve from data and experience

Deep Learning

Machine learning with large neural networks

Generative AI

Deep learning that creates text, images, code

Programming vs. machine learning



Example: instead of writing spam rules by hand, show the computer 10,000 emails labelled “spam” / “not spam”.

Three ways machines learn



Supervised

Learn from examples with answers

Business: Predict which customers will leave



Unsupervised

Find patterns without answers

Business: Group customers into segments



Reinforcement

Learn by trial, error and reward

Business: Set prices that maximise revenue over time

Narrow AI vs. general AI



Narrow AI

Great at one task

- Chess engines
- Spam filters
- Face recognition

Exists today — everywhere



General AI (AGI)

Human-level at any task

- Learns anything
- Transfers knowledge
- Common sense

Does not exist yet — hotly debated

AI borrows from many fields



Philosophy

How does the mind reason?



Mathematics

Logic, probability, computation



Economics

How to decide to maximise payoff?



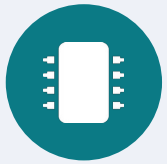
Neuroscience

How does the brain process information?



Psychology

How do people think and act?



Computer engineering

How to build fast computers?



Control theory

How can machines control themselves?



Linguistics

How does language relate to thought?

How did we get here?

70 years of hype, disappointment and breakthroughs

The birth of AI (1943–1973)

From big dreams to the first disappointment



First artificial neuron

McCulloch & Pitts model a neuron as a simple on/off unit

Turing's paper

"Computing Machinery and Intelligence" proposes the imitation game

AI gets its name

Dartmouth workshop: "artificial intelligence" is born

A dose of reality

Translation fails, problems too big — funding is cut

Rules, winters and data (1970s–2010)

From hand-written knowledge to learning from data



Expert systems

Human expertise written as if-then rules (e.g. medical diagnosis)

AI winter

Expert systems prove brittle; companies and funding collapse

Neural nets return

Backpropagation makes multi-layer networks trainable

Probability & data

Learning from data replaces hand-written rules; big data arrives

The deep learning era (2011–today)

Data + computing power + neural networks



ImageNet moment

Deep networks beat all other image-recognition methods

AlphaGo

Beats a world champion at Go, a game long thought too hard

Transformers

A new architecture — the basis of today's language models

ChatGPT

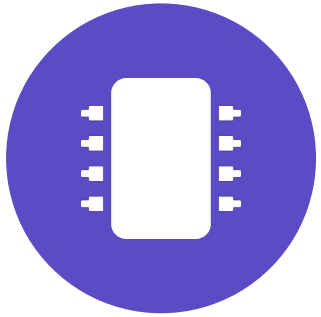
Generative AI reaches the public; business adoption explodes

Why is AI booming now?



Data

The internet, phones and sensors produce huge labelled datasets



Computing power

GPUs and the cloud make training huge models affordable



Better algorithms

Deep learning and transformers turn data into skill

Most ideas are decades old. Data and computing caught up.

What AI can do today



Self-driving vehicles



Translation



Speech recognition



Recommendations



Game playing



Image recognition



Medical diagnosis



Writing text, code & images

AI across the business



Marketing

Customer segments, personalised offers, churn prediction



Finance

Fraud detection, credit scoring, forecasting



Operations

Demand forecasting, inventory, route planning



HR

CV screening, workforce planning



Customer service

Chatbots, sentiment analysis of reviews



Strategy

Market analysis, decision support

Benefits and risks



Benefits

- More productive work!
- Faster medical diagnosis
- Scientific discovery
- Less repetitive, dangerous work



Risks

- Biased decisions (hiring, credit)
- Surveillance & privacy
- Job displacement
- Misinformation & deepfakes
- Autonomous weapons

Key takeaways

- AI = machines that act well in new situations
- Most AI today: rational agents
- AI ⊃ Machine Learning ⊃ Deep Learning ⊃ Gen AI
- Progress came in waves: hype → winter → new idea